AI Applications in Research

A Concept Summary, with Emphasis on Claude Code
for Agentic, Reproducible Empirical Work

Prepared as a scheduled research digest

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Abstract

This document summarizes how AI—and **agentic coding assistants** such as **Claude Code** in particular—is being applied across the research lifecycle: literature work, data analysis, replication, writing, and peer review. It distills a practical philosophy that has emerged among quantitative social scientists (notably in the applied econometrics community around Scott Cunningham's "Mixtape" work): treat the model as a *thinking partner* rather than an autonomous author, and let AI raise the standard of *reproducibility and adversarial verification* rather than merely accelerate output.

1 Why AI Now Matters for Research

The recent shift is not chatbots that answer questions but **agents** that operate inside the researcher's own environment—reading files, running code, editing manuscripts, and calling tools across sessions. This moves AI from an *advice* role to an *execution* role, which changes both the opportunities and the risks:

- **Opportunity:** compress the distance between an idea and a tested result, and make rigorous practices (cross-checks, documentation, audits) cheap enough to always do.
- **Risk:** plausible-but-wrong output at scale—silent bugs, fabricated citations, and unexamined assumptions—if the workflow lacks verification discipline.

2 A Guiding Philosophy

A workable stance for research use of tools like Claude Code rests on a few principles:

1. **Thinking partner, not autopilot.** The model executes when the next move is clear and surfaces ambiguity when it is not; the *dialogue itself* is part of the intellectual work, not just a delivery mechanism.
2. **Design before results.** Get the specification right (identification, estimand, sample) before interpreting any estimate.
3. **Separate author from auditor.** Adversarial review must run in a fresh session with no prior commitments; the auditor never edits the author's code.
4. **Documentation is a primary output.** Undocumented work is effectively incomplete; persistent notes counter a model's session "amnesia."
5. **Formal process over intuition.** Checklists and structured protocols beat informal judgement, especially for error-prone, repetitive verification.

3 Applications Across the Research Lifecycle

3.1 Reading and synthesis

Long PDFs are split into small chunks and deep-read with structured note extraction, turning passive reading into a persistent, queryable markdown record of claims, methods, and gaps.

3.2 Data analysis and estimation

The assistant scaffolds and runs analysis pipelines, but the decisive practice is **cross-language replication**: the same estimand is computed independently in R, Stata, and Python and required to agree to many decimal places. Divergence flags a bug in specification, data handling, or a package default—caught before it reaches a table.

3.3 Reproducibility and code audits

A multi-audit "referee" protocol (e.g., code review, cross-language replication, directory structure, output automation, and econometric soundness) is run by a *fresh* model instance that inspects but never modifies the author's code—mirroring an external referee.

3.4 Bibliography and citation integrity

Many narrow-focus agents verify each reference against canonical sources, directly targeting the well-known failure mode of confident but fabricated or mis-attributed citations.

3.5 Writing, figures, and presentations

End-to-end generation of slides and figures (e.g., Beamer/TikZ) with automated visual checks (collision/overlap passes, zero-warning compiles) and rhetoric audits that push from narrative framing toward technical precision.

3.6 Peer review

AI agents assist a human referee—organizing, stress-testing, and checking claims—while the *referee report itself remains human-written*, keeping accountability with the reviewer.

4 Illustrative Toolkit

The table sketches representative Claude Code "skills" for empirical research, as a concrete instance of the applications above.

Skill	What it does
Referee 2	Five-audit reproducibility protocol run by a fresh instance; inspects, never edits.
Blindspot	"Peripheral vision" audit that surfaces overlooked problems and unasked questions.
Bibcheck	Parallel per-citation verification against canonical sources.
Split-PDF	Chunked deep reading of papers with structured, persistent notes.
Beautiful Deck	Beamer generation with audience triage plus rhetoric and graphics audits.
GTD Harness	Warrant-first tracking of hypotheses, evidence, and falsification tests.

5 Verification: The Core Discipline

Because generative systems produce fluent errors, AI-assisted research depends on making verification *routine and cheap*:

- **Triangulate** results across independent implementations and, where possible, trust visual inspection of pictures over raw numbers.
- **Isolate** the auditor from the author so review is genuinely adversarial.
- **Persist** context (decisions, warrants, open questions) in version-controlled markdown so work is recoverable and falsifiable.
- **Automate** the checklist so rigor does not depend on a busy researcher's memory.

6 Limitations and Cautions

- **Hallucination and silent bugs** remain the central threat—verification is not optional.
- **Provenance and reproducibility:** record prompts, tool versions, and seeds so an AI-assisted pipeline can be re-run by others.
- **Authorship and accountability:** the researcher owns the claims; AI drafts and checks, but does not sign the referee report or the paper.
- **Privacy and policy:** respect data-use agreements and journal/venue rules on AI assistance and disclosure.

7 Takeaways

1. Agentic assistants shift AI from advice to *execution* inside the research environment, changing both speed and risk.
2. The payoff comes not from faster typing but from making *rigorous, reproducible, adversarially verified* practice the default.
3. Used as a disciplined thinking partner—with cross-checks, isolated audits, and persistent documentation—tools like Claude Code can raise, rather than lower, the standard of empirical work.

Selected references: Scott Cunningham, *Scott's Mixtape* Substack (https://causalinf.substack.com); *MixtapeTools* repository (https://github.com/scunning1975/mixtapetools); Anthropic, *Claude Code* documentation (https://docs.anthropic.com/en/docs/claude-code/overview).